Week 4 Summary – Data Visualization with Matplotlib and Seaborn

Objective

The objective of Week 4 was to learn the basics of data visualization using **Matplotlib** and **Seaborn**. The focus was on creating different types of charts and graphs to analyze and present data in a clear and meaningful way.

Concepts Learned

- Learned the importance of data visualization in Data Science.
- Understood how to use **Matplotlib** for creating line plots, bar charts, histograms, pie charts, and scatter plots.
- Learned how to use **Seaborn** to create advanced visualizations such as heatmaps and pairplots.
- Added titles, axis labels, and legends to charts for better readability.
- Understood how visualizations help identify trends, patterns, and relationships in data.

Hands-on Programs Completed

1. Line Plot
2. Bar Chart
3. Histogram
4. Pie Chart
5. Scatter Plot
6. Heatmap using Seaborn
7. Pairplot using Seaborn
8. Box Plot using Seaborn

Client Project

Project Title: Sales Dashboard

Project Description

Developed a Sales Dashboard using **Pandas**, **Matplotlib**, and **Seaborn** to visualize sales data. The project displays monthly sales and profit through different charts, including line plots, bar charts, scatter plots, and heatmaps. These visualizations help analyze sales performance and understand the relationship between sales and profit.

Skills Used

- Pandas DataFrame
- Matplotlib
- Seaborn

- Line Plot
- Bar Chart
- Scatter Plot
- Heatmap
- Data Visualization
- Data Analysis

Learning Outcome

This week improved my understanding of data visualization techniques using Python. I learned how to present data graphically, identify trends and patterns, and create simple dashboards for data analysis. These skills are essential for communicating insights effectively in Data Science and Machine Learning projects.